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Eight supplementary demand/band calibration tasks: Construction

Contents and concept route



Classify after putting the equation in standard form

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Classify after putting the equation in standard form

Concept difficulty: Medium concept

1 Classify after putting the equation in standard form

Medium concept

Section 1 of 8

Next →

Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

1 Collect the variable terms.

Why valid: Adding $2x$ to both complete sides preserves equality.

Result: $5x - 4 = 1$

2 Remove the constant.

Why valid: Adding 4 to both sides is reversible.

Result: $5x = 5$

3 Select the category.

Why valid: The coefficient 5 is nonzero, so exactly one rational can satisfy the equation.

Result: One solution.

4 Challenge an all-solutions claim.

Why valid: One counterexample suffices to refute that every rational works.

Result: $x=0$ gives -4 and 1 , which differ.

Requested output: category of solution set

$$3x - 4 = 1 - 2x$$

No solution | One solution | Every rational

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

The key step

The first has every rational; the second has none. The reduced constant comparison matters when the coefficient is zero.

Worked explanation

Over the rationals, classify $3x - 4 = 1 - 2x$ as having no solution, one solution, or every rational as a solution. Use the criterion $Ax = B$: A nonzero gives one rational solution, $A = B = 0$ gives every rational, and $A = 0$ with B nonzero gives none. Do not report the value of x .

Follow the operation, its justification and its result.

1. **Collect the variable terms.**

Why valid: Adding $2x$ to both complete sides preserves equality.

Result: $5x - 4 = 1$

2. **Remove the constant.**

Why valid: Adding 4 to both sides is reversible.

Result: $5x = 5$

3. **Select the category.**

Why valid: The coefficient 5 is nonzero, so exactly one rational can satisfy the equation.

Result: One solution.

4. **Challenge an all-solutions claim.**

Why valid: One counterexample suffices to refute that every rational works.

Result: $x=0$ gives -4 and 1 , which differ.

Result

One solution.

Check

At $x=0$ the two sides differ, so the category is not every rational. The nonzero coefficient rules out none.

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1-2-3 QUICK CHECK

1 · CHECK

At $x=0$ the two sides differ, so the category is not every rational. The nonzero coefficient rules out none.

Now you do one

Classify $4x+1=4x+1$ and $4x+1=4x+2$. Do not solve by dividing by the cancelled coefficient.

Classify linked parameter exceptions

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Classify linked parameter exceptions

Concept difficulty: Medium concept

1 Classify linked parameter exceptions

Medium concept

Section 2 of 8

← Previous

Next →

Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

1 Expand E1 and cancel 2x.

Why valid: Distribution and subtracting the same term from both sides preserve the solution set.

Result: E1 becomes $-6=k$.

2 Classify E1.

Why valid: The reduced statement has no x , so it holds for all x or for none.

Result: Every rational if $k = -6$; none otherwise.

3 Factor the E2 constant.

Why valid: $4 - 2k = -2(k - 2)$.

Result: $(k - 2)x = -2(k - 2)$.

4 Separate $k=2$ before division.

Why valid: At $k=2$ both sides vanish; division by $k-2$ would lose this case.

Result: E2 has every rational if $k=2$.

5 Classify the remaining k .

Why valid: A nonzero rational coefficient has a unique rational quotient.

Result: E2 has one rational solution for $k \neq 2$.

6 Cross-check the exceptional parameters.

Why valid: Each category must be evaluated in the original pair.

Result: At $k = -6$: E1 every, E2 one. At $k = 2$: E1 none, E2 every.

k is fixed; x ranges over rationals

$$E1: 2(x-3) = 2x+k$$

$$E2: (k-2)x = 4-2k$$

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

The key step

At $t=3$ every rational works; at $t=4$ exactly one works. Split a zero divisor before cancelling.

Worked explanation

Let k be rational and x rational. For each k , classify each of these equations as having none, one, or every rational solution: E1: $2(x-3)=2x+k$; E2: $(k-2)x=4-2k$. Use the $Ax=B$ criteria; a numerical solution is not requested.

Follow the operation, its justification and its result.

1. **Expand E1 and cancel $2x$.**

Why valid: Distribution and subtracting the same term from both sides preserve the solution set.

Result: E1 becomes $-6=k$.

2. **Classify E1.**

Why valid: The reduced statement has no x , so it holds for all x or for none.

Result: Every rational if $k=-6$; none otherwise.

3. **Factor the E2 constant.**

Why valid: $4-2k=-2(k-2)$.

Result: $(k-2)x=-2(k-2)$.

4. **Separate $k=2$ before division.**

Why valid: At $k=2$ both sides vanish; division by $k-2$ would lose this case.

Result: E2 has every rational if $k=2$.

5. **Classify the remaining k .**

Why valid: A nonzero rational coefficient has a unique rational quotient.

Result: E2 has one rational solution for $k \neq 2$.

6. **Cross-check the exceptional parameters.**

Why valid: Each category must be evaluated in the original pair.

Result: At $k = -6$: E1 every, E2 one. At $k = 2$: E1 none, E2 every.

Result

E1: every rational at $k = -6$, none otherwise. E2: every rational at $k = 2$, one otherwise.

Check

Substitute $k = -6$ and $k = 2$ in both originals. Neither equation has a hidden division step at its exceptional parameter.

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1-2-3 QUICK CHECK

1 · CHECK

Substitute $k = -6$ and $k = 2$ in both originals. Neither equation has a hidden division step at its exceptional parameter.

Now you do one

Classify $(t-3)y=2(t-3)$ at $t=3$ and $t=4$, keeping the exceptional case.

Classify with a coupled domain boundary

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Classify with a coupled domain boundary

Concept difficulty: Medium concept

1 Classify with a coupled domain boundary

Medium concept

Section 3 of 8

← Previous

Next →

Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

1 Exclude the parameter making the second equation undefined.

Why valid: Its right denominator $k-2$ must be nonzero.

Result: $k=2$ is INVALID PARAMETER.

2 Isolate the zero-factor case $k=1$.

Why valid: The first equation then reads $0=0$, while the second remains defined.

Result: The second becomes $(x+1)/(x-1)=-1$, giving $x=0$; it is rational and allowed.

3 Reduce the first equation for k outside $\{1,2\}$.

Why valid: Only in this branch is cancellation of $k-1$ followed by division by $k-2$ valid.

Result: $x=1/(k-2)$.

4 Reduce the second equation with both denominator conditions retained.

Why valid: Cross-multiplication is reversible when $x \neq 1$ and $k \neq 2$.

Result: $(k-2)(x+1)=k(x-1)$, hence $x=k-1$.

5 Require the two candidates to coincide.

Why valid: Both original equations must hold for the same x .

Result: $(k-1)(k-2)=1$, hence $k^2-3k+1=0$.

6 Use the rational-domain restriction.

Why valid: The only compatibility parameters $(3 \pm \sqrt{5})/2$ are irrational; $x=k-1$ is then irrational and excluded.

Result: No rational solution for every k outside $\{1,2\}$.

7 Verify the exceptional allowed branch in the originals.

Why valid: A reduced-line result is insufficient when divisions were used in other branches.

Result: $k=1, x=0$ gives $0=0$ and $1/(-1)=1/(-1)$. ONE at $k=1$; INVALID at $k=2$; NONE otherwise; never EVERY.

k real; x rational; both equations required

$$(k-1)(k-2)x = k-1$$

$$(x+1)/(x-1) = k/(k-2); \text{ denominators nonzero}$$

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

The key step

At $a=2$ the equation is defined but inconsistent (no solution); at $a=3$ exactly one rational works; at $a=2+\sqrt{2}$ the sole real candidate $1/\sqrt{2}$ is irrational, so no rational works. Domain checks come after valid branch reduction.

Worked explanation

Extension beyond routine linear equations. k is real, x must be rational. Classify the simultaneous system by k as INVALID PARAMETER, NO RATIONAL SOLUTION, ONE RATIONAL SOLUTION, or EVERY RATIONAL SOLUTION: $(k-1)(k-2)x=k-1$ and $(x+1)/(x-1)=k/(k-2)$. An equation with a zero denominator is invalid; a solution must satisfy both original equations and $x \neq 1$. Report categories and the parameter cases, not a general numerical solution method.

Follow the operation, its justification and its result.

1. **Exclude the parameter making the second equation undefined.**

Why valid: Its right denominator $k-2$ must be nonzero.

Result: $k=2$ is INVALID PARAMETER.

2. **Isolate the zero-factor case $k=1$.**

Why valid: The first equation then reads $0=0$, while the second remains defined.

Result: The second becomes $(x+1)/(x-1)=-1$, giving $x=0$; it is rational and allowed.

3. **Reduce the first equation for k outside $\{1,2\}$.**

Why valid: Only in this branch is cancellation of $k-1$ followed by division by $k-2$ valid.

Result: $x=1/(k-2)$.

4. **Reduce the second equation with both denominator conditions retained.**

Why valid: Cross-multiplication is reversible when $x \neq 1$ and $k \neq 2$.

Result: $(k-2)(x+1)=k(x-1)$, hence $x=k-1$.

5. **Require the two candidates to coincide.**

Why valid: Both original equations must hold for the same x .

Result: $(k-1)(k-2)=1$, hence $k^2-3k+1=0$.

6. **Use the rational-domain restriction.**

Why valid: The only compatibility parameters $(3 \pm \sqrt{5})/2$ are irrational; $x=k-1$ is then irrational and excluded.

Result: No rational solution for every k outside $\{1,2\}$.

7. Verify the exceptional allowed branch in the originals.

Why valid: A reduced-line result is insufficient when divisions were used in other branches.

Result: $k=1, x=0$ gives $0=0$ and $1/(-1)=1/(-1)$. ONE at $k=1$; INVALID at $k=2$; NONE otherwise; never EVERY.

Result

ONE RATIONAL SOLUTION at $k=1$; INVALID PARAMETER at $k=2$; NO RATIONAL SOLUTION for every other real k ; never EVERY.

Check

At $k=1$, $x=0$ satisfies both originals. At $k=2$ a denominator is zero. At the two real compatibility roots, x is irrational, so a real solution does not meet the rational-domain request.

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1-2-3 QUICK CHECK

1 • CHECK

At $k=1$, $x=0$ satisfies both originals. At $k=2$ a denominator is zero. At the two real compatibility roots, x is irrational, so a real solution does not meet the rational-domain request.

Now you do one

Let y be rational and a real. Classify $(a-2)y=1$ at $a=2$, $a=3$ and $a=2+\sqrt{2}$. Keep undefined operations separate from empty solution sets.

Execute an inverse chain across a table and a process graph

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Execute an inverse chain across a table and a process graph

Concept difficulty: Medium concept

1 Execute an inverse chain across a table and a process graph

Medium concept

Section 4 of 8

← Previous

Next →

Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

1 Convert the exact observation.

Why valid: Ten millimetres are one centimetre.

Result: $C=22/10=11/5$ centimetres.

2 Undo $C=(B+2)/5$.

Why valid: Multiplication by nonzero 5 and subtraction of 2 reverse this stage.

Result: $B=5C-2=9$.

3 Undo $B=3A+4$.

Why valid: Subtract 4, then divide by nonzero 3.

Result: $A=(9-4)/3=5/3$.

4 Undo $A=(X-1)/2$.

Why valid: Multiply by nonzero 2, then add 1.

Result: $X=2(5/3)+1=13/3$ centimetres.

5 Run the entire forward chain.

Why valid: A correct recovered input must reproduce the original observation after every named operation.

Result: $13/3 \rightarrow 5/3 \rightarrow 9 \rightarrow 11/5$ centimetres = 22 millimetres.

Observation table: $C = 22$ mm; program uses cm

$X \rightarrow$ subtract 1, divide 2 $\rightarrow A$

$A \rightarrow$ multiply 3, add 4 $\rightarrow B \rightarrow$ add 2, divide 5 $\rightarrow C$

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

The key step

These are the same observation. Both give $B=8$, $A=4/3$ and $X=11/3$ cm. Different results reveal a unit mapping error.

Worked explanation

A measuring program uses numerical readings in centimetres: $X \rightarrow A=(X-1)/2 \rightarrow B=3A+4 \rightarrow C=(B+2)/5$. The observation table gives $C=22$ millimetres. The inverse procedure is prescribed: convert the observation to centimetres, undo the C operation, then B , then A . Execute it to recover X and verify every forward stage. Treat the displayed readings as exact.

Follow the operation, its justification and its result.

1. **Convert the exact observation.**

Why valid: Ten millimetres are one centimetre.

Result: $C=22/10=11/5$ centimetres.

2. **Undo $C=(B+2)/5$.**

Why valid: Multiplication by nonzero 5 and subtraction of 2 reverse this stage.

Result: $B=5C-2=9$.

3. **Undo $B=3A+4$.**

Why valid: Subtract 4, then divide by nonzero 3.

Result: $A=(9-4)/3=5/3$.

4. **Undo $A=(X-1)/2$.**

Why valid: Multiply by nonzero 2, then add 1.

Result: $X=2(5/3)+1=13/3$ centimetres.

5. **Run the entire forward chain.**

Why valid: A correct recovered input must reproduce the original observation after every named operation.

Result: $13/3 \rightarrow 5/3 \rightarrow 9 \rightarrow 11/5$ centimetres = 22 millimetres.

Result

$X=13/3$ centimetres.

Check

The forward chain gives $A=5/3$, $B=9$, $C=11/5$ cm =22 mm, so neither a unit mistake nor a reversed-step mistake survives the check.

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1-2-3 QUICK CHECK

1 · CHECK

The forward chain gives $A=5/3$, $B=9$, $C=11/5$ cm =22 mm, so neither a unit mistake nor a reversed-step mistake survives the check.

Now you do one

Run the same inverse program with $C=20$ millimetres, and separately with $C=2$ centimetres. Should X differ?

Choose the governing equation from its conditions

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Choose the governing equation from its conditions

Concept difficulty: Medium concept

1 Choose the governing equation from its conditions

Medium concept

Section 5 of 8

← Previous

Next →

Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

1 Read the coefficient condition.

Why valid: The problem explicitly excludes $A=0$.

Result: A is nonzero.

2 Select R1.

Why valid: Division by the fixed nonzero rational A is defined and reversible.

Result: Use $x=B/A$.

3 Check the alternative boundary.

Why valid: The all-solutions rule needs $A=B=0$ and therefore conflicts with the given $A \neq 0$.

Result: R2 does not apply.

Given: $Ax = B$; $A \neq 0$

Candidate R1: divide by A

Candidate R2: zero-coefficient identity

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

The key step

The first admits every rational, the second none. A rule is selected by both coefficients, not the word linear.

Worked explanation

A linear calibration is stated as $Ax=B$ with rational A and B , and A is known to be nonzero. Choose the governing rule for recovering x : R1, $x=B/A$; R2, infer every rational is a solution from $0=0$. Only choose the applicable rule; do not compute a value.

Follow the operation, its justification and its result.

1. **Read the coefficient condition.**

Why valid: The problem explicitly excludes $A=0$.

Result: A is nonzero.

2. **Select R1.**

Why valid: Division by the fixed nonzero rational A is defined and reversible.

Result: Use $x=B/A$.

3. **Check the alternative boundary.**

Why valid: The all-solutions rule needs $A=B=0$ and therefore conflicts with the given $A \neq 0$.

Result: R2 does not apply.

Result

R1, $x=B/A$.

Check

The boundary $A=B=0$ would admit R2, showing why the stated nonzero condition determines this choice.

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1-2-3 QUICK CHECK

1 · CHECK

The boundary $A=B=0$ would admit R2, showing why the stated nonzero condition determines this choice.

Now you do one

Choose the rule when $A=0, B=0$, and when $A=0, B=1$.

Translate an ordered pair without changing meaning

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Translate an ordered pair without changing meaning

Concept difficulty: Medium concept

1 Translate an ordered pair without changing meaning

Medium concept

Section 6 of 8

← Previous

Next →

Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

1 Map the table headings to coordinate slots.

Why valid: The convention gives the first slot to x and the second to y .

Result: Slots are (x,y) .

2 Insert the two labelled values.

Why valid: The row assigns $x=2$ and $y=3$.

Result: $(2,3)$.

3 Check a swapped pair.

Why valid: Its first coordinate would claim $x=3$, contrary to the row.

Result: (3,2) represents a different point.

Table heading: x | y

Row: 2 | 3

Coordinate template: (horizontal, vertical)

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

The key step

They give (4,1) and (1,4), distinct points. Coordinate roles are part of the meaning.

Worked explanation

The ordered-pair table has columns x and y and one row $x=2$, $y=3$. Write that row as a point in Cartesian ordered-pair notation. The coordinate convention is horizontal x first and vertical y second; no equation solving is requested.

Follow the operation, its justification and its result.

1. **Map the table headings to coordinate slots.**

Why valid: The convention gives the first slot to x and the second to y .

Result: Slots are (x,y) .

2. **Insert the two labelled values.**

Why valid: The row assigns $x=2$ and $y=3$.

Result: $(2,3)$.

3. **Check a swapped pair.**

Why valid: Its first coordinate would claim $x=3$, contrary to the row.

Result: $(3,2)$ represents a different point.

Result

$(2,3)$.

Check

Reading the pair back gives $x=2, y=3$, preserving both table headings.

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1-2-3 QUICK CHECK

1 • CHECK

Reading the pair back gives $x=2, y=3$, preserving both table headings.

Now you do one

Translate rows $x=4, y=1$ and $x=1, y=4$. Are their points identical?

Order two dependent inverse steps

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Order two dependent inverse steps

Concept difficulty: Medium concept

1 Order two dependent inverse steps

Medium concept

Section 7 of 8

← Previous

Next →

Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

1 Read the forward dependency.

Why valid: The doubled quantity already includes the added 3.

Result: Start $\rightarrow$ add 3 $\rightarrow$ double $\rightarrow$ output.

2 Reverse the outer operation first.

Why valid: Halving recovers the quantity before doubling, which still contains the added 3.

Result: Halve first, then subtract 3.

3 Check the competing order on a changed start.

Why valid: An independent forward/reverse check tests the dependency rather than memorising a rule.

Result: Start 2 gives output 10; halve then subtract gives 2, subtract then halve gives 3.5.

Start → add 3 → double → 14

Inverse components: halve | subtract 3

Required: dependency-respecting order

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

The key step

The first needs subtract 3 then halve; the second halve then subtract 3. The component list is unchanged, but the dependency changes.

Worked explanation

A starting number is first increased by 3 and then doubled, producing 14. You know the inverse components: subtract 3, and halve. Coordinate them into the correct reverse order. State which component must come first and why; do not calculate the starting number.

Follow the operation, its justification and its result.

1. **Read the forward dependency.**

Why valid: The doubled quantity already includes the added 3.

Result: Start $\rightarrow$ add 3 $\rightarrow$ double $\rightarrow$ output.

2. **Reverse the outer operation first.**

Why valid: Halving recovers the quantity before doubling, which still contains the added 3.

Result: Halve first, then subtract 3.

3. **Check the competing order on a changed start.**

Why valid: An independent forward/reverse check tests the dependency rather than memorising a rule.

Result: Start 2 gives output 10; halve then subtract gives 2, subtract then halve gives 3.5.

Result

Halve first, then subtract 3; undo the last operation before its input operation.

Check

For a changed start 2, the forward output is 10; the selected inverse returns 2 while the competing order returns 3.5.

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1-2-3 QUICK CHECK

1 • CHECK

For a changed start 2, the forward output is 10; the selected inverse returns 2 while the competing order returns 3.5.

Now you do one

For forward double then add 3, which inverse comes first? Compare with forward add 3 then double.

Give the warrant for a reversible division

Subject: Mathematics

Topic: Algebraic decisions and domain boundaries

Concept: Give the warrant for a reversible division

Concept difficulty: Medium concept

1 Give the warrant for a reversible division

Medium concept

Section 8 of 8

← Previous

Conditions for this construction

- Use the exact declared domains and the stated category/operation conventions.

1 State the divisor property.

Why valid: The given divisor 2 is nonzero.

Result: Division by 2 is defined.

2 Connect the operation to its inverse.

Why valid: Multiplying both sides back by 2 restores the same equality.

Result: The step is reversible and preserves the solution set, establishing $x=4$.

3 Challenge the zero-divisor boundary.

Why valid: Division by zero is undefined, so it has no reversible operation of this kind.

Result: The warrant cannot license division by zero.

Premise: $2x = 8$

Claim: $x = 4$

Requested: a warrant for the division step

Separate the declared givens and representation roles from the protected decision.

Stage 1 of 2

Align declared givens, domains and requested output while preserving the decisive learner work.

The key step

It becomes $0=0$ and admits every real, so the non-reversible step enlarged the solution set. Identical operations need the relevant reversibility condition.

Worked explanation

Over the reals, an author concludes $x=4$ from $2x=8$ by dividing both sides by 2. Supply one warrant explaining why the conclusion follows and why the same justification would fail for division by zero. No further equation solving is requested.

Follow the operation, its justification and its result.

1. **State the divisor property.**

Why valid: The given divisor 2 is nonzero.

Result: Division by 2 is defined.

2. **Connect the operation to its inverse.**

Why valid: Multiplying both sides back by 2 restores the same equality.

Result: The step is reversible and preserves the solution set, establishing $x=4$.

3. **Challenge the zero-divisor boundary.**

Why valid: Division by zero is undefined, so it has no reversible operation of this kind.

Result: The warrant cannot license division by zero.

Result

Because 2 is nonzero, dividing both sides by 2 is defined and reversible, preserving equality and the solution set. Division by zero is undefined.

Check

Multiplication by 2 takes $x=4$ back to $2x=8$. The zero-divisor case lacks this reversible division.

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1-2-3 QUICK CHECK

1 • CHECK

Multiplication by 2 takes $x=4$ back to $2x=8$. The zero-divisor case lacks this reversible division.

Now you do one

From $x=1$ multiply both sides by zero. Does the resulting equation have exactly the original solution?

Mathematics · Eight supplementary demand/band calibration tasks